SHEET # 2

2.1 Prove that the DFT, X (k), of a sequence x (n) is periodic.

2.2 Calculate the DFT of the data sequence {0, 1, 1, 0} and check the validity of your answer by calculating its IDFT.

2.3 Use the results of prob. (2.2) to verify Parseval's theorem for the data {0,1,1,0}.

2.4 Calculate the DFT of the data sequence {0,1,1,0} using the decimation-in-time FFT algorithm. Check the answer with that of Prob. (2.2). Compare numbers of complex additions and multiplications in the two methods. Calculate the IFFT to verify your answer.

2.5 Calculate the FFT of the data sequence {0,0,1,1,1,1,0,0} and plot the amplitude and phase spectra. Check your answer by calculating the IFFT to obtain the original data sequence.

2.6 For the following signal, $X(n) = \cos(\frac{n\pi}{2})$, find the first 8-samples and then using the FFT algorithm, calculate the frequency components, X(k). Assume we use the DFT algorithm instead of the FFT algorithm, what is the computational saving.

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